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Opposing Effects of Impurity Ion Sr^{2+} on the Heterogeneous Nucleation and Growth of Barite ($BaSO_4$)

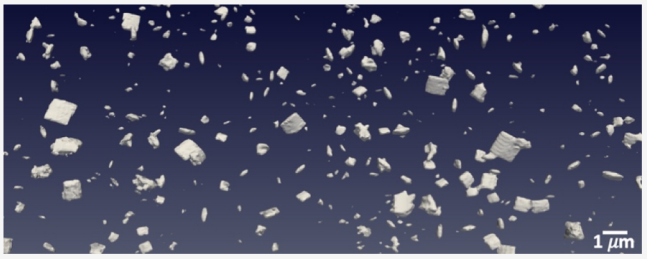
Ke Yuan,* Vitalii Starchenko, Nikhil Rampal, Fengchang Yang, Xiaogang Yang, Xianghui Xiao, Wah-Keat Lee, and Andrew G. Stack*

Cite This: *Cryst. Growth Des.* 2021, 21, 5828–5839

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ABSTRACT: Nucleation of barite ($BaSO_4$) has broad implications in geological, environmental, and materials sciences. While impurity metals are common, our understanding of how they impact nucleation remains dim. Here, we used classical optical microscopy compared to fast X-ray nanotomography (XnT) to investigate heterogeneous nucleation of barite on silica in situ with Sr^{2+} as an impurity ion. The observed barite nucleation rates were consistent with classical nucleation theory (CNT), where barite crystals displayed a nonuniform size distribution, exhibiting distinct morphologies and incubation periods in Sr-free solutions. While undetectable with optical microscopy, nanotomography revealed that addition of Sr^{2+} enhances nucleation rates driven by the prefactor in CNT, likely because both adsorbed Ba^{2+} and Sr^{2+} act as precursor sites on which nucleation occurs. Sr^{2+} simultaneously inhibits growth, however, leading to a homogeneous distribution of smaller crystals. This finding will enable an improved predictive understanding of nucleation in natural and synthetic environments.



INTRODUCTION

Nucleation and growth of crystals at solid−water interfaces are key processes in ore/mineral deposition, heavy metal sequestration, and materials synthesis.[1,2] For example, the sparingly soluble mineral barite ($BaSO_4$) is formed from aqueous barium (Ba^{2+}) and sulfate (SO_4^{2-}) ions, which are relatively abundant and widely distributed in the near subsurface. Barite commonly forms as scale during oil and gas operations,[3] and in materials science, it is used as an additive in lead acid batteries that facilitates lead sulfate nucleation during discharge.[4] In environmental science, crystallization of barite can incorporate impurity metal contaminants, such as Ra^{2+} and Sr^{2+}, due to their chemical similarity with barium.[5,6] Because of this, dissolution of Ra-bearing barite found in uranium mines is the critical factor controlling Ra^{2+} mobility.[7] Similarly, barite found in seawater that is undersaturated with respect to that phase is intimately linked to the presence of Sr^{2+} (i.e., "barite paradox"), where the Sr isotopes in barite could be a useful indicator for paleo-ocean environments.[8] It is known that impurities such as Sr^{2+} poison the advance of monomolecular steps observed on calcite and barite crystals under atomic force microscopy.[9−12] Barite forms solid solutions with the end member celestite ($SrSO_4$), and previous studies have resulted in a theory that strives to account for nucleation of ideal $Ba_xSr_{(1-x)}SO_4$ solid solutions.[13−15] However, theoretical predictions such as these on ideal systems have not been sufficiently validated by experimental results. For example, surface processes are known to play a central role in barite precipitation such as barite formed on organic surfaces of living organisms and in confined pores.[16,17] Despite its importance, the mechanisms by which the common impurity ion, Sr^{2+}, impacts interfacial processes leading to the heterogeneous nucleation and growth of barite remain poorly understood.

There has been increasing evidence that nucleation of carbonates, phosphates, and iron oxides can occur via nonclassical nucleation pathways, where crystals are not nucleated by individual ion attachment but form through a multistep reaction process with the presence of intermediate metastable phases and/or particle aggregation (e.g., dense liquid phases, prenucleation clusters, and polymorphs).[1,2] However, nucleation of other minerals (e.g., barite) under certain saturation states can be explained by the classical nucleation theory (CNT), and even each individual step in multiple-step nonclassical nucleation can be treated with CNT successfully.[18] Unlike calcite, which has multiple crystalline polymorphs and may contain several intermediate phases, barite has no known polymorphs. This suggests its nucleation pathway(s) might be more limited than that of calcite. However, recent studies

Received: June 21, 2021
Revised: August 23, 2021
Published: September 8, 2021



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https://doi.org/10.1021/acs.cgd.1c00715
Cryst. Growth Des. 2021, 21, 5828−5839

suggesting amorphous barite formation raise new questions on whether barite nucleation follows CNT or not.[19−21]

We focus on the heterogeneous nucleation and growth of barite at silica−water interfaces, where the influence of Sr^{2+} on both nucleation and growth of barite was investigated. Numerous methods, including optical/electron microscopy, atomic force microscopy, and X-ray scattering, have been developed to study crystal nucleation.[22−26] Optical microscopy has been routinely used in crystal nucleation studies by recording the number of crystals in a flow cell.[27] Since nuclei need to grow into a large enough size to be resolved by the optical light, which often have a practical resolution close to 1 μm (ideal resolution from 200 to 350 nm estimated as half the wavelength). Early-stage nucleation events and nuclei smaller than the resolution limit cannot be taken into account as nuclei are much smaller than the practical resolution. Liquid-cell transmission electron microscopy (TEM) can image crystal nucleation from the very beginning, with a very high resolution, but the limited field of view and changes in chemistry due to the creation of radical species by the beam is a potential problem. Similar to TEM, the limited field of view of atomic force microscopy cannot yield sufficient crystal count statistics to analyze the density of nuclei on a substrate robustly.[28,29] Small angle X-ray scattering (SAXS) and grazing incidence SAXS (GISAXS) are increasingly used methods (operating in the reciprocal space) to obtain total scattering volume and particle sizes. Particle size commonly observed in (GI)SAXS is between 1 and 50 nm, but once crystals grow into micron size ranges, they will not be "seen" by the X-ray beam in the small angle range.[30−32] While this valuable technique has expanded our understanding considerably, experiments are often limited in terms of the ranges of saturation index (SI) and/or nucleation density that can be explored and have the potential to underestimate nuclei density when large particles form rapidly in the system. Furthermore, for GISAXS, quantification of the total number of nuclei relies on knowing the effective cross-sectional area of the beam on the substrate, which is often difficult to obtain accurately.

Here, we used a different approach, employing fast X-ray nanotomography (XnT) based on transmission X-ray microscope (TXM), to image the nucleation and growth of barite in situ with a field of view of 40 μm × 40 μm and a spatial resolution of 30 nm.[33] The fast XnT scans in less than 1 min and provides sufficient temporal resolution to catch the rapid crystal growth process and to minimize radiation damages to the samples.[34] The 3D tomography provides the total number and mass of nuclei in the system, which yields the nucleation and growth rates, respectively. The tomography data also provides spatial distribution of crystals on the substrate, allowing growth rates from individual crystals of different orientations to be compared. Comparing XnT to conventional optical microscopy, we found that optical microscopy results provide reasonable estimates of the interfacial energy but underestimate the pre-factor term in CNT due to its poor resolution. By adding Sr^{2+}, we observe significant changes in the crystal morphology and a reduction in growth rate, but the nucleation rate is found to increase. In Sr-free solutions, a relatively higher growth rate is observed but with a lower nucleation rate. In addition, crystals with a high-index crystal surface attached to the substrate appear earlier on the surface and develop into micron-sized crystals compared with barite attached via the (001) surface, which has a longer incubation time and appears as small crystals. Tomography images were processed using traditional and automated segmentation methods, developed based on a convolutional neural network (CNN). Combined, we found Sr^{2+} suppresses barite growth while promoting nucleation, specifically by increasing the pre-factor in CNT instead of the common attribution that are driven by a reduction in interfacial energy. The counteracting effect of Sr^{2+} on barite nucleation and growth may be relevant to the micron-sized Sr-bearing barite observed in marine sediments and could have general implications on the role of cation impurities played during nucleation and growth of other solid solution minerals.

METHODS

Solutions and Flow Cell Systems. Solutions containing $BaCl_2$ (Sigma-Aldrich), $SrCl_2$(Sigma-Aldrich), and Na_2SO_4 (Sigma-Aldrich) were prepared by dissolving the salts into deionized water. The saturation index (SI = $\log([Ba^{2+}][SO_4^{2-}]/K_{sp,barite}$, $K_{sp,barite} = 10^{-9.98}$) of solutions used for barite nucleation is SI = 1.9, 2.1, and 2.5. Another set of Sr-rich solutions with Ba:Sr = 1:1 were prepared under similar SI with respect to pure barite (see supporting information, Table S1 for solution compositions). The flow cell consisted of two solution reservoirs in glass bottles containing the $BaCl_2$/$BaCl_2$ + $SrCl_2$ and Na_2SO_4 (whose concentrations were determined by the SI with respect to barite), a syringe pump set at a flow rate of 100 μL/min (Cole-Parmer, model 270), a custom solution mixer, and a glass capillary cell that served as a reaction vessel, where nucleation was observed. The mixer is a sealed glass bottle of a maximum volume of 2.0 mL with two inlets from the stock solutions and one outlet, whose lines were made of Teflon tubing (1/32 in. in inner diameter). Some studies have shown that at SI > 2.0, homogeneous nucleation of barite may occur,[35,36] but in other studies, homogeneous nucleation was not found at SI 2.1.[37] To minimize nucleation prior to the portion of the capillary that was imaged, the solution mixer was placed close to the inlet of the capillary, such that the volume of the mixed solution prior to being exposed to the capillary was about 0.5 mL and with a residence time of 2.5 min under flowing solution. We did not observe any signs of significant homogeneous nucleation during our experiments, which could lead to pressure increase in the mixer (indicated by an increase of solution volume) due to the homogeneously formed crystals blocking the outlet of the tube. A magnetic stir bar inside the solution mixer mixed the solution using a stir plate (Fisher Scientific, model Lab Disc S55). The mixer and attached tubes were thoroughly cleaned in ultrasound water bath before each use. Square borosilicate tubes (500 μm × 500 μm inner diameter, VitroCom) were used for the optical microscopy measurements. Rounded quartz capillary tubes (100 μm to 150 μm inner diameter, Charles Supper Company) were used for the XnT. The glass tubes were soaked in piranha solutions overnight and washed with deionized water before use to remove any oils or films left from their manufacture. The thin capillaries were carefully mounted onto a specially designed XnT cell where solution flowed from the bottom and drained simultaneously on the top without interrupting the stage rotation (Figure S1, details on the flow cell will be reported separately). The thin capillaries were washed with 1 M HCl and deionized water after being mounted on the XnT stage before flowing the growth solutions.

Optical Microscopy. The heterogeneous nucleation process of barite on the square glass tube was observed in situ using an optical microscope (Olympus BX51) equipped with a digital camera and an image recording system. The field of view is 700 μm × 300 μm under a magnification of 300×. The image was taken every 5 min.

Fast X-ray Nanotomography. XnT was performed with the TXM at the full-field X-ray imaging beamline (18-ID) of National Synchrotron Light Source II, Brookhaven National Laboratory. The TXM was operated at 9keV and had a field of view of 40 μm × 40

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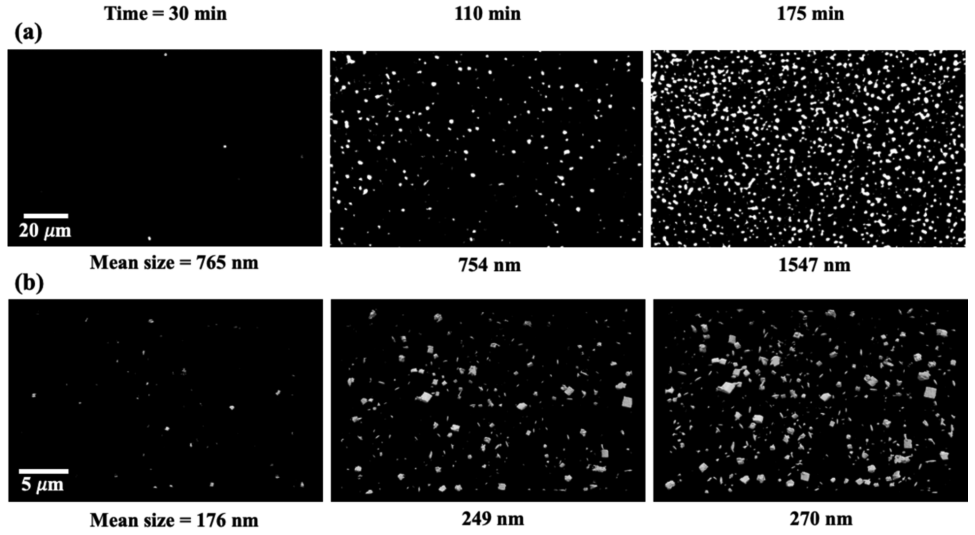


Figure 1. Images showing nucleation of barite in SI = 2.47, Sr-free solutions. (a) Optical microscopy (Binary images are shown to clearly illustrate the distribution and size of particles) and (b) XnT images collected at t = 30, 110, and 175 min. Notice the difference in the scale bar.

μm and a resolution of 30 nm. The sample was imaged in situ in a flow cell with an exposure time of 0.05 s for each projection and a total acquisition time of about 1 min for a full 3D tomography scan. To minimize radiation artifacts, each 1 min XnT scan was followed by a 10 min rest during which no beam was exposed to the sample. Upon finishing the time-dependent nucleation studies at each SI, we performed three additional scans at a location upstream to the observed position in the flow direction (to avoid potential radical damage imposed by previous scans) to obtain experimental errors on the measured nuclei density (see Supporting Information, Error Analysis). The XnT data were processed and reconstructed in TomoPy.[38−40] Volume and surface area were calculated from the segmented images in Fiji ImageJ.[41−43] A surface resampling value within the Particle Analysis Toolbox in Fiji ImageJ[42] was used to reduce the noise and remove crystals below the resolution limit when counting the particle density (see Supporting Information for details). Volumes of the crystals were calculated based on the number of voxels within the segmented images, and masses of the precipitates were calculated based on a barite density of 4.48 g/cm^3. In some cases, automatic segmentation of grayscale images was used, where the traditional thresholding method was challenging due to increasing noise in the dataset. Recently, machine-learning-based image segmentation has been proved efficient in such challenging cases. In this work, we employed a convolutional neural network (CNN)-based segmentation method.[44] A small portion (1%) of the grayscale images and the corresponding manual segmentation images were used for training the network. The trained network was then applied to segment all the grayscale images.

RESULTS

Nucleation Kinetics of Barite Obtained by Optical Microscopy and XnT. We compared the barite nucleation kinetics obtained from optical microspecies and XnT under the same solutions. An increasing number of crystals in Sr-free (SI = 2.47) solutions were observed over time on both optical and XnT images (Figure 1). Under optical light, larger crystals (>1 μm) exhibited a darker color compared to smaller ones (Figure S2), and both were segmented as shown in the binary images (Figure 1a). Average particle size ranges between 0.8 μm and 1.5 μm in optical microscopy, whereas average particle size is about 0.2 μm with fine resolution under XnT. At 175 min, significant numbers of crystals appeared to grow together as imaged by optical microscopy (Figure 1a). The particle overlap became more significant at higher SI, where the nucleation density was underestimated. It is reflected as the plateau on the counted nucleation density at 200 min as shown in Figure 2a,b (highlighted by the boxes, data not used for fitting the nucleation rates). Significant growth was observed after 100 min in Sr-free solutions under optical light; however, the 100 min apparent incubation time reduced to less than 25 min as revealed by XnT (Figure 2a,c). Nucleation density measured by optical microscopy is one order of magnitude lower than that measured by XnT (Figure 2). With the addition of Sr^{2+}, optical microscopy showed fewer crystals than in Sr-free solutions, whereas for XnT, more crystals were revealed in Sr-rich solutions. This is because, in the presence of Sr^{2+}, reduced particle size is observed. From the time dependency of the nucleation density, the linear part of the curves can be used to obtain a slope as the nucleation rate. Nucleation rates were found to be 2 times higher on average when Sr^{2+} was added, as revealed by XnT, whereas a contrary conclusion was obtained by optical light due to its low resolution in resolving small particles (Table 1).

We used CNT originally derived for homogeneous nucleation to fit the data, and it was plotted as ln(J) vs 1/SI^2 in Figure 2e. The fitted slope and intercept corresponding to interfacial energy and pre-factor, respectively, based on eq 1.

$$J = \Gamma \exp\left(\frac{-\Delta G^*}{kT}\right)$$

$$\Delta G^* = \frac{B\sigma^3 V_{\mathrm{mol}}^2}{(2.3SI \times kT)^2}$$

$$\ln J = \ln \Gamma + \left(\frac{-B\sigma^3 V_{\mathrm{mol}}^2}{2.3^2 (kT)^3}\right) \times \frac{1}{SI^2} \quad (1)$$

Where J is the nucleation rate in nuclei/m^2/s, B is the geometric factor (16.76, treated as a spherical particle to compare optical with XnT results), σ is the interfacial energy in J/m^2 (for heterogeneous nucleation, this is an apparent, or effective interfacial energy reflecting the energies of the substrate, precipitate and solution), V_{mol} is the molar volume

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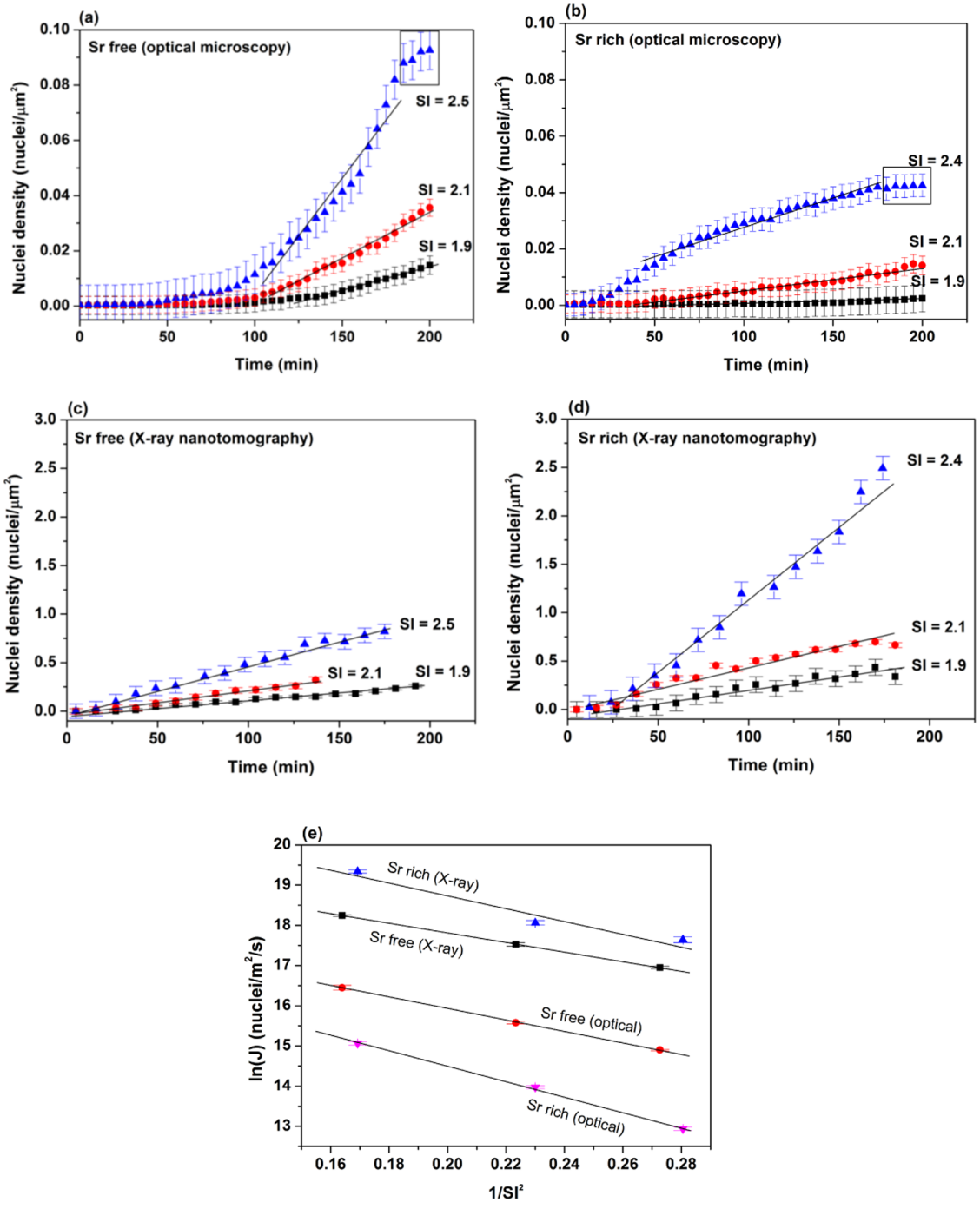


Figure 2. Nuclei density of barite changes with time in (a, c) Sr-free and (b, d) Sr-rich solutions of three different SI. Results obtained by (a, b) optical microscope were compared with those acquired by (c, d) XnT. The areas indicated by the rectangles indicate where nuclei density measurements become unreliable using optical microscopy due to an inability to resolve individual particles. (e) $\ln(J)$ vs $1/SI^2$ plot for the calculations of pre-factor and interfacial energy in the classical nucleation theory.

of barite (52.11 cm^3/mol), SI is the saturation index of the solution, k is the Boltzmann constant, T is the temperature in K, and Γ is the pre-factor term accounting for ion diffusion, desolvation, and other secondary processes at the solid−water interface.

Interfacial energies showed a slight increase by about 10% in Sr-rich solutions, as shown in results measured by both optical and X-ray microscopy (Table 2). Even though the absolute number of nuclei on the substrate may be underestimated by optical microscopy, the slope which represents the gradient of nuclei density increase with time can still be fitted with reasonable accuracy by using optical microscopy (Figure 2e). On the other hand, the pre-factor term (intercept of lines) is underestimated by the optical microscopy, in that the results obtained by X-ray microscopy yielded a higher pre-factor than that obtained by optical microscopy (Table 2).

Growth Kinetics of Barite Crystals. Volume data obtained from segmented tomography images was used to calculate the volume of barite crystals. To obtain the growth

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Table 1. Nucleation Rates of Barite Determined by Using Optical Microscopy and XnT

	optical microscopy	XnT
SI (Sr-free)	nucleation rate $\times 10^6$ (nuclei/m^2/s)	nucleation rate $\times 10^7$ (nuclei/m^2/s)
1.92	2.96 ± 0.06	2.30 ± 0.08
2.12	5.83 ± 0.19	4.08 ± 0.17
2.47	13.89 ± 0.79	8.37 ± 0.21
SI (Sr-rich)	nucleation rate $\times 10^6$ (nuclei/m^2/s)	nucleation rate $\times 10^7$ (nuclei/m^2/s)
1.89	0.41 ± 0.02	4.60 ± 0.33
2.09	1.17 ± 0.04	7.00 ± 0.41
2.43	3.48 ± 0.15	25.10 ± 1.04

Table 2. Fitted Pre-Factor $\ln \Gamma$ and Interfacial Energy (σ) by Optical Microscopy (Optical) and XnT (X-ray) for the Heterogeneous Nucleation of Barite

samples	$\ln \Gamma$ (nuclei/m^2/s)	σ (mJ/m^2)
Sr-free (optical)	18.72 ± 0.01	34.53 ± 0.03
Sr-rich (optical)	18.31 ± 0.05	38.23 ± 0.15
Sr-free (X-ray)	20.19 ± 0.02	32.67 ± 0.09
Sr-rich (X-ray)	22.08 ± 0.71	36.42 ± 2.48

rate of the crystals in the commonly used units of mol/m^2/s, here, two types of surface areas, substrate surface area and crystal surface area, were used. Crystal surface areas are the ideal measurement for computing the growth rate, but as we show, the substrate surface area is a more robust measurement, and the derived growth rate is less prone to uncertainty. Here, we demonstrated that these two methods of calculating the surface area provided different perspectives in understanding the barite growth kinetics. When the substrate surface area was used, which is the underneath area of the silica glass imaged by XnT, the growth rate increased with increasing SI in both Sr-free and Sr-rich solutions (Figure 3a,b). The fitted slopes in Figure 3 indicated that the growth rates are similar in Sr-free and Sr-rich solutions of the same SI (Table 3), that is, the fraction of barite on the substrate grows at a similar rate no matter if Sr^{2+} was added or not. However, when surface areas of the crystals were calculated, growth rates were reduced by about half in solutions with Sr^{2+} (Figure 4 and Table 3). Crystal surface area was estimated by summing the geometric surface areas of all the individual crystals, which were

Table 3. Growth Rates of Barite Determined by Using XnT

	substrate surface area	crystal surface area
SI (Sr-free)	growth rate $\times 10^{-8}$ (mol/m^2/s)	growth rate $\times 10^{-8}$ (mol/m^2/s)
1.92	3.37 ± 0.17	9.21 ± 0.71
2.12	4.40 ± 0.35	9.44 ± 0.99
2.47	7.90 ± 0.30	9.46 ± 0.89
SI (Sr-rich)	growth rate $\times 10^{-8}$ (mol/m^2/s)	growth rate $\times 10^{-8}$ (mol/m^2/s)
1.89	2.85 ± 0.20	5.90 ± 0.28
2.09	4.38 ± 0.22	6.13 ± 0.27
2.43	8.37 ± 0.48	5.41 ± 0.48

calculated from the number of voxels occupied by barite, with an approximation that all crystals exhibited a cubic shape. The surface areas per gram of barite (specific surface areas) were higher in all Sr-rich solutions, corresponding to a small particle size (Figure 4a). The crystal surface area-based growth rates do not show statistically significant variations with SI (Figure 4b), which likely arises because the (significant) uncertainty introduced by estimating the geometric surface area is larger than the variation with SI. We also tried to directly calculate the crystal surface area from the tomography data using surface meshes enclosing all crystals. Both manually segmented data and machine learning-based automated segmentation results were compared (Figure S3). This estimation has high uncertainty for small crystals due to challenges in creating reasonable surface mesh when the surface to mass ratios were high (Figure S4 and Table S2). Growth rates obtained by both geometric-based and directly-calculated surface areas showed a reduction in Sr-rich solutions compared to that in Sr-free solutions.

To fit our substrate surface area-based growth rate, we used a simplified linear rate law (eq 2) derived from transition state theory.

$$r = k\left(\frac{Q}{K_{sp}} - 1\right) \tag{2}$$

where r is the growth rate of barite in mol/m^2/s, K_{sp} is the solubility product of barite = $10^{-9.98}$, Q is the ion activity product of $a_{Ba^{2+}} \cdot a_{SO_4^{2-}}$, k is the forward rate constant. From

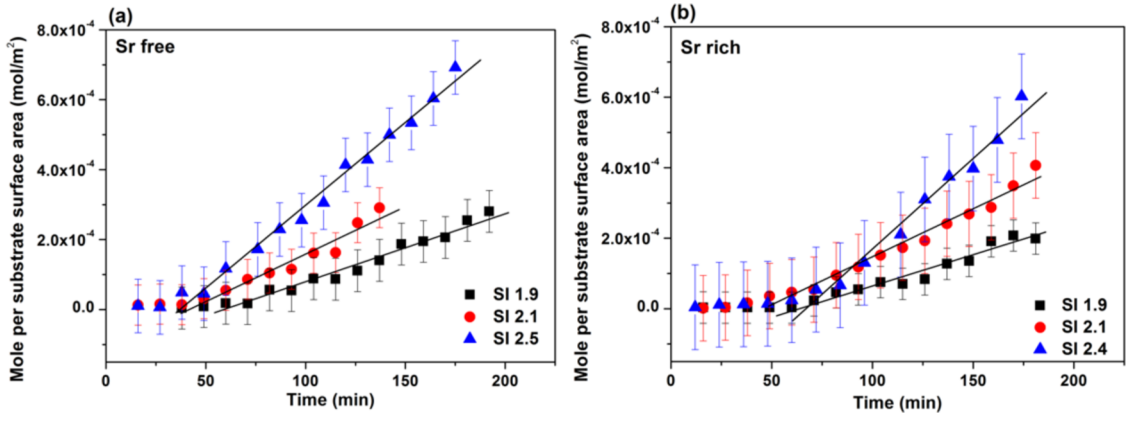


Figure 3. Substrate surface area-based growth rates of barite measured by XnT in (a) Sr-free and (b) Sr-rich solutions.

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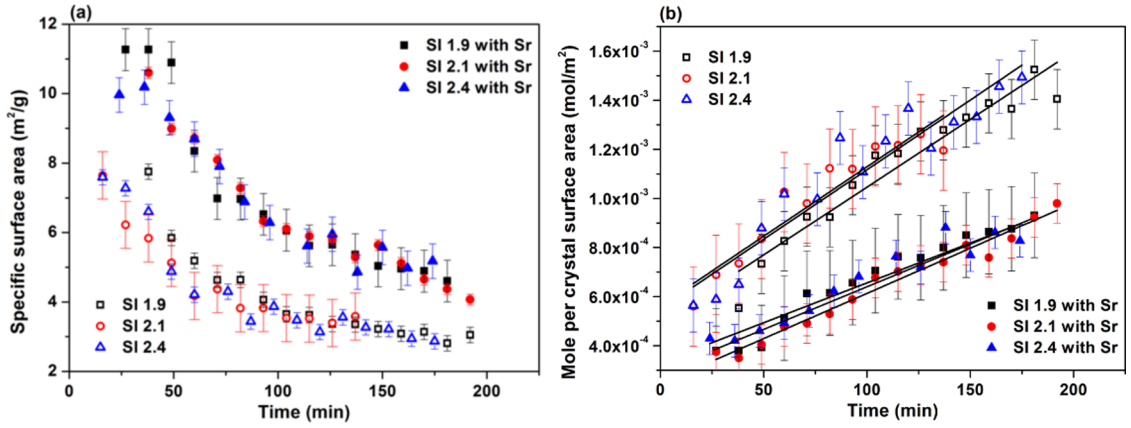


Figure 4. Changes of the crystals (a) specific surface areas and (b) geometric crystal surface area-based growth rates calculated in Sr-free (hollow symbols) and Sr-rich solutions (solid symbols).

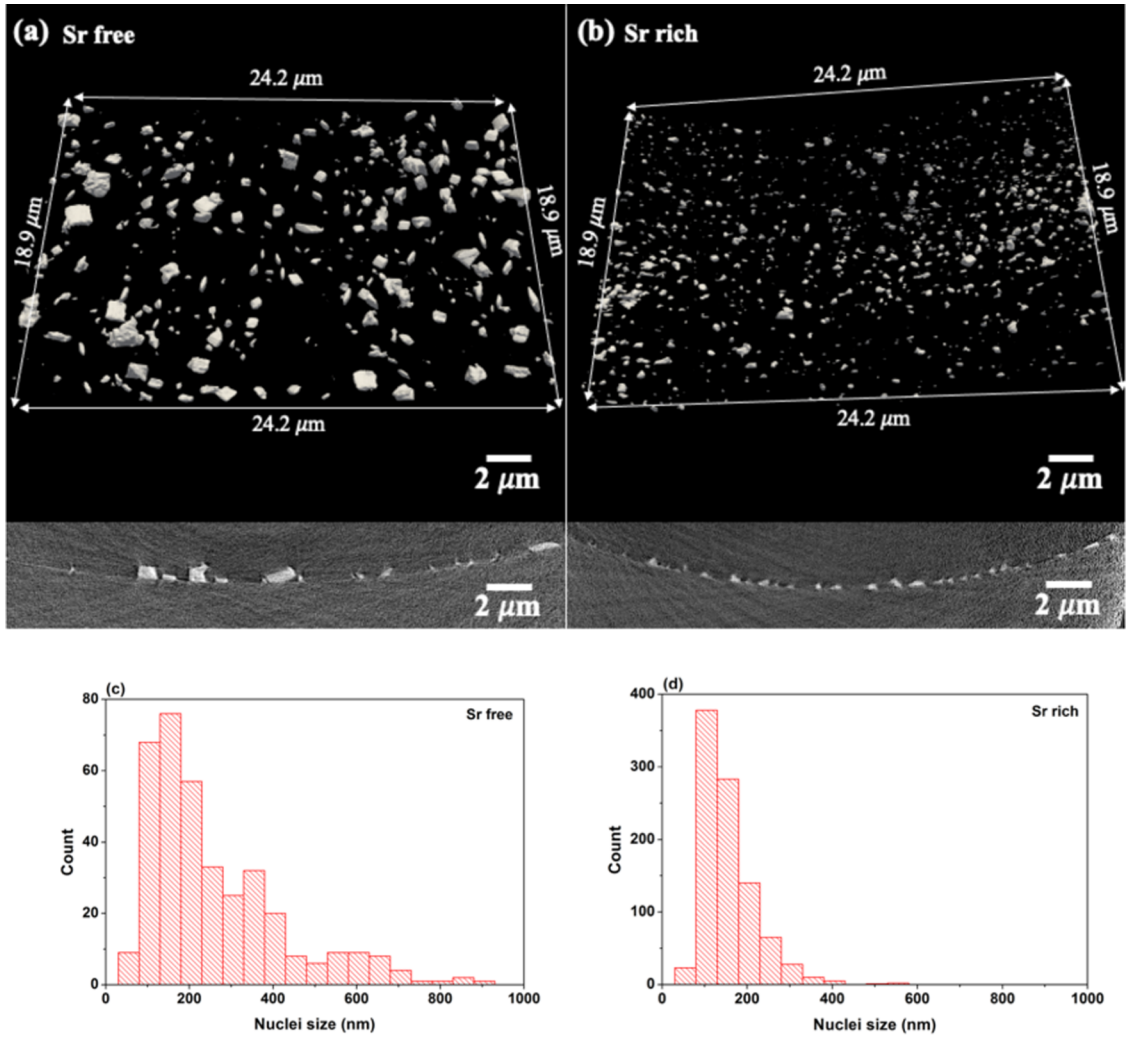


Figure 5. Three-dimensional view of barite crystal morphology in (a) Sr-free and (b) Sr-rich solutions with SI = 2.47 at 175 min. Cross-sectional views of the nanotomography data are shown below. Corresponding particle size distributions are shown in (c) and (d).

the $(Q/K_{sp} - 1)$ vs r plot, we obtained the rate constants in Sr-free and Sr-rich solutions from the slopes, which showed the average value of $2.47 \pm 0.50 \times 10^{-10}$ mol/m^2/s. We also analyzed growth rates obtained by optical microscopy, which were up to 1 order of magnitude higher at the highest SI = 2.4 than that calculated by XnT (Figure S5 and Table S3). We attribute this discrepancy to the defocused edges of the crystals under optical light, which leads to an overestimation of the surface coverage of the crystals on the substrate.

Homogenous Crystal Distributions in Sr-Rich Solutions. As mentioned previously, the nucleation density in Sr-rich solutions was underestimated using optical micros-

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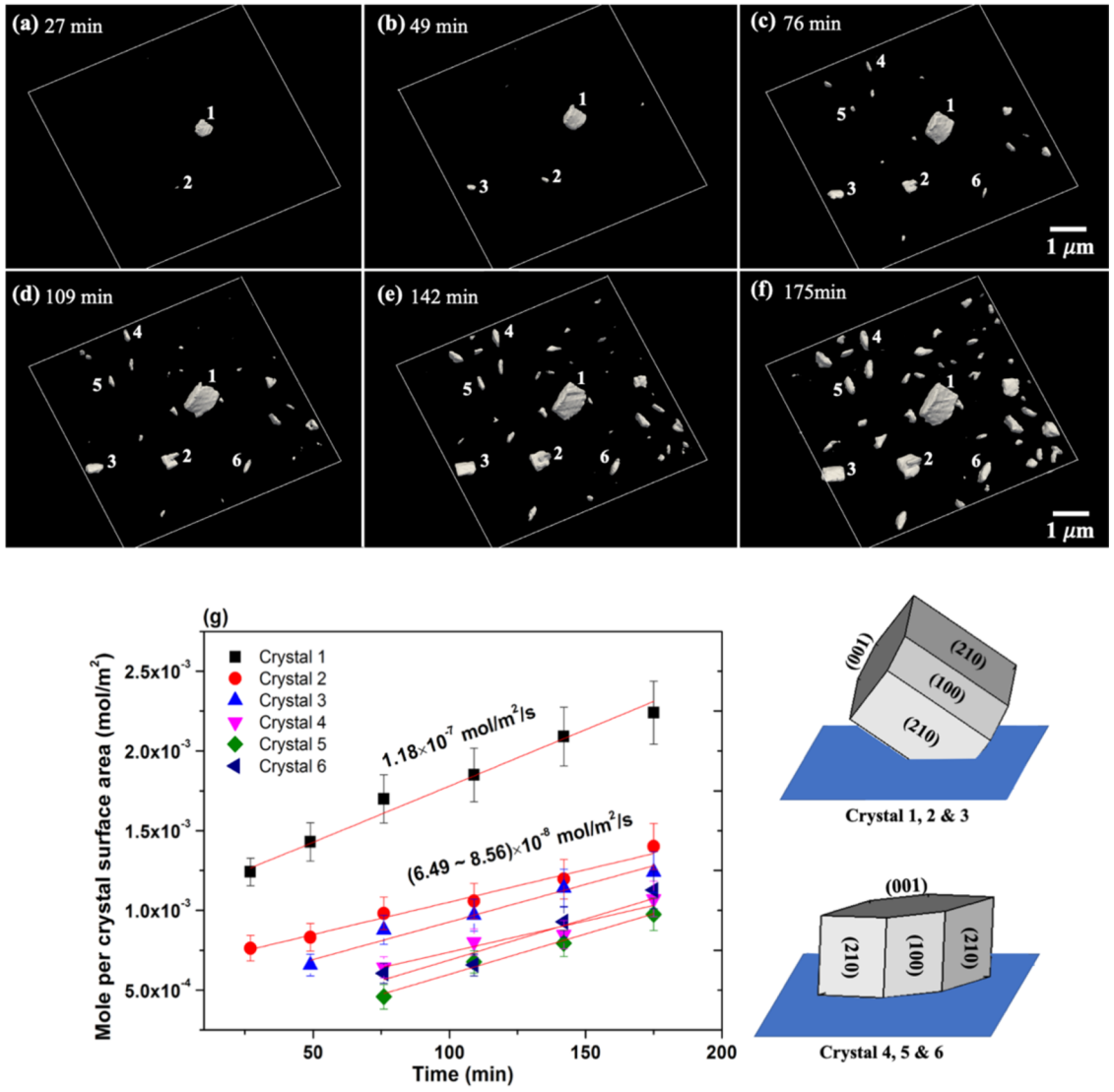


Figure 6. Zoomed-in view (7.1 μm × 6.3 μm) of barite growth in Sr-free (SI = 2.43) solutions as a function of time (a–f). (g) Growth kinetics of six individual crystals as labeled in (a) to (f). Pictures on the left show orientations of two types of barite crystals.

copy because the crystals were smaller than the practical resolution limit, while waiting for longer times led to an inability to resolve individual crystals (Figure 2). In solutions of SI = 2.47 at t = 175 min, the majority of the particles are between 100 and 200 nm in both Sr-free and Sr-rich solutions (Figure 5). However, a significant number of large particles having sizes exceeding 400 nm were found in Sr-free solutions (Figure 5a,c) but not in Sr-rich solutions (Figure 5b,d). As shown before in Figure 3, adding Sr^{2+} did not change the substrate surface area-based estimate of the growth rate of barite but changed how barite particles were distributed on the substrate. That is, in addition to being smaller due to a lower growth rate, their sizes were more uniformly distributed. This results in a narrower size distribution relative to the Sr-free case. A similar trend was observed in two other solutions at lower SI, whereas the effects from strontium became more obvious with increasing SI (Figure S6).

Orientation-Dependent Incubation Time of Barite Crystals. Because crystals in Sr-free solutions at SI = 2.4 grow into large sizes with visible crystal facets, we isolated and aligned part of the data to show the evolution of individual crystals with time (Figure 6.). We tracked six crystals in total, with crystals 1, 2, and 3 having a rhombic shape with a high-index surface attached to the substrate, and their (210) and (100) surfaces are the major facets visible in Figure 6. Crystals 4, 5, and 6 all have their (001) surface attached to the substrate instead, appearing as needle-like crystals. Interestingly, crystals 1, 2, and 3 had a short incubation time and appeared first on the substrate at times earlier than 49 min. All other crystals, including 4, 5, and 6, were observed at later times. All six crystals had a similar growth rate in the range of (6.49–11.8) × 10^{-8} mol/m^2/s (Figure 6g), which is consistent with the global growth rate (9.46 × 10^{-8} mol/m^2/s) shown in Table 3.

Effects of (Ba, Sr)SO_4 Solid Solution on Nucleation Kinetics. If the precipitates are not pure barite but a mixed $(Ba_xSr_{1-x})SO_4$ solid solution phase, we need to consider the effects of the solid solution on the nucleation rate. Based on Pina 2001 et al.,[13] the composition (x) of the solid solution will impact the nucleation rate (J). Equation 1 can be rewritten as:

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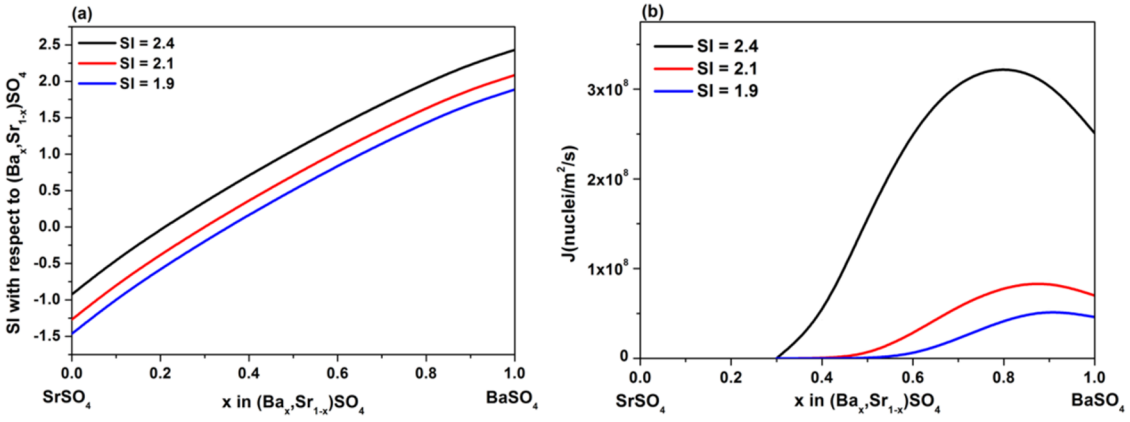


Figure 7. Calculated (a) supersaturation and (b) nucleation rates of $(Ba_x,Sr_{1-x})SO_4$ solid solutions at SI (with respect to pure barite) of 2.4, 2.1, and 1.9 in Sr-rich solutions.

$$J(x) = \Gamma(x)\exp\left[\frac{-B\sigma^3(x)V_{mol}^2(x)}{k^3T^3(\ln S(x))^2}\right] \quad (3)$$

where S (supersaturation index), σ (interfacial energy), V_{mol} (molar volume), and Γ (pre-factor) are not constants but a function of composition (x) of the solid. These parameters all increase with the increase of x (Figure S7).

The pre-factor Γ can be expressed in eq 4:[45]

$$\Gamma(x) = \sqrt{\frac{\sigma(x)}{kT}}\left(\frac{kT}{hd_0^2}\right)\exp\left(\frac{-\Delta G_{diff}}{kT}\right) \quad (4)$$

where h is Planck's constant, and d_0 is the size of the nuclei (here, we use d_0 = 1 nm as an approximation), ΔG_{diff} is the activation free energy for transferring a structural unit (ions) from the solution to the nucleus. Here, ΔG_{diff} was assumed to be constant and was back-calculated based on eqs 3 and 4 using our measured nucleation rates in Sr-rich solutions with XnT (Table 1). The calculated ΔG_{diff} was 170 kJ/mol when precipitating a pure barite phase. To identify the effect of this energy difference on nucleation rates, we calculated the theoretical supersaturation of solid solutions and their nucleation rates J as a function of x (Figure 7). The curves appeared in similar shapes with maximum nucleation rates at x = 0.8−0.9, which were about 1.2 to 1.3 times higher than that of pure barite (x = 1). Of note, the maximum nucleation rates (Figure 7b) do not correspond to the highest supersaturation values (Figure 7a). These calculations predict that precipitation of a slightly Sr-rich barite (x = 0.8−0.9 in $(Ba_x,Sr_{1-x})SO_4$) should have a higher nucleation rate than precipitating a pure barite phase.

DISCUSSION

Sr^{2+}-Promoted Barite Nucleation. Previous studies have suggested that aqueous divalent ions may induce a higher nucleation rate of barite by lowering the surface free energy.[46] The interfacial energy of pristine barite in supersaturated solutions has been estimated to be about 76−105 mJ/m^2 during homogeneous nucleation.[47] Our results, however, indicate Sr^{2+} mainly enhances nucleation rates by increasing the kinetic pre-factor term (Γ), without lowering the effective interfacial energy of nucleation, 36.4 ± 2.5 mJ/m^2. (Table 1 shows the strontium slightly increases the interfacial energy, which would suppress the nucleation rate. See SI, Figure S8). The pre-factor is often poorly defined,[2] and can be influenced by, e.g., desolvation and diffusion of Ba^{2+} and SO_4^{2-} ions during nucleation, kinetic limits on the aggregation of smaller particles to a large cluster, or by the efficiency of ion collisions with the mineral surface. The slope, related to interfacial energy, is well-characterized by both optical microscopy and X-ray tomography (Table 1), but often no changes are detected in the pre-factor with optical microscopy because of the limited resolution.[27] Thus, an effective interfacial energy is often fitted to the experimental nucleation data when applying CNT. Here, we showed that the pre-factor can be a critical term in explaining our observation, but it was resolvable only when high-resolution XnT was applied.

When considering the effect of the strontium on the nucleation rate, it is important to first consider the formation of $(Ba,Sr)SO_4$ solid solutions that will likely change the thermodynamics of precipitation. If solid solutions formed in our Sr-rich solutions of SI = 2.4, the predicted nucleation rate would be the highest when precipitating a $(Ba_{0.8}Sr_{0.2})SO_4$ phase, where its nucleation rate is 1.3 times faster than forming a pure barite phase (Figure 7). This prediction is rationalized by Nielson et al.,[48] who observed a general inverse relation between solubility and surface energy among different crystals, where minerals with low solubilities tend to have higher surface energies in an aqueous solution. Celestine is more soluble than barite;[3] therefore, Sr-containing barite is more soluble and should have a low interfacial energy compared to pure barite. This reduction in interfacial energy favors the nucleation of a $(Ba_{0.8}Sr_{0.2})SO_4$ phase over pure barite as predicted by theory (Figure 7). In contrast to the theory prediction, however, our measured nucleation rates in Sr-rich solutions are more than doubled (2.2 times higher on average) than that in Sr-free solutions (Table 1), instead of a 30% increase as expected by solid solution effects. When $(Ba,Sr)SO_4$ solid solutions are formed, the interfacial energy will decrease as predicted by theory (Figure S7b). However, our data did not show a decrease in interfacial energies by adding Sr^{2+} (Table 1). Thus, even though nucleation of a slightly Sr-rich barite phase is likely more favorable thermodynamically, $(Ba,Sr)SO_4$ solid solution formation effects should result in a much smaller increase in nucleation rates in the Sr-rich solution than what is observed here.

To consider other possible ways that strontium may affect the nucleation of barite, we postulate that a likely initial step in the nucleation process that strongly influences the rate is

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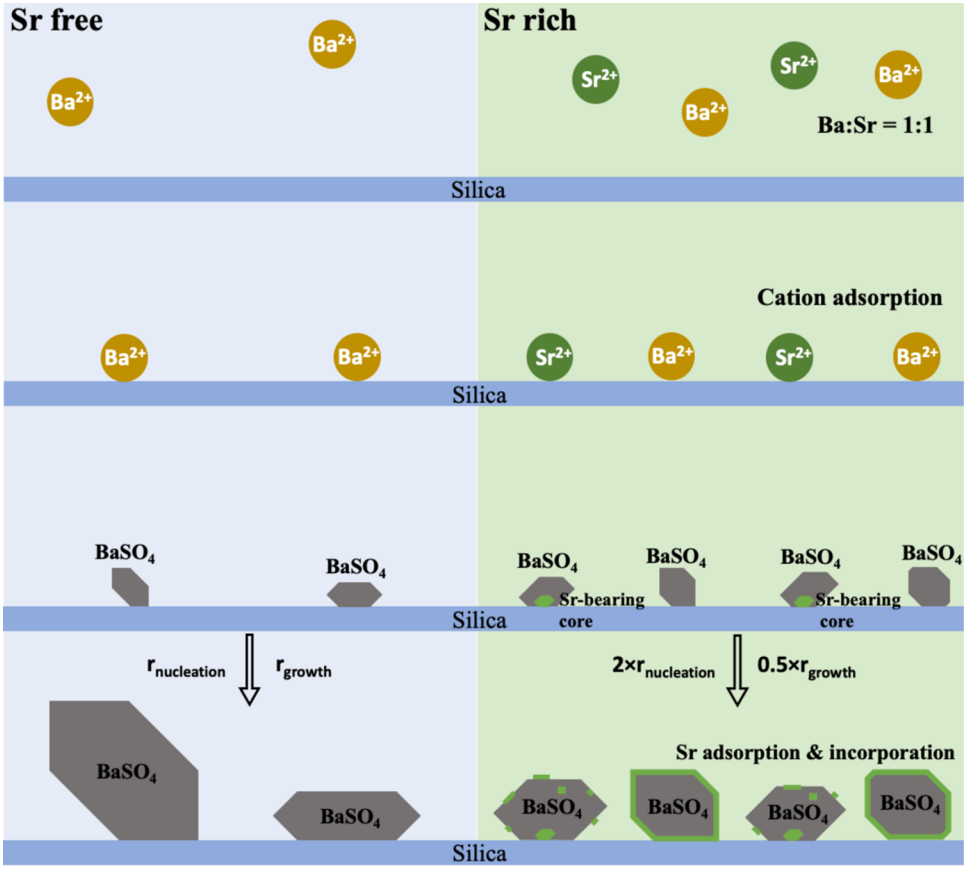


Figure 8. Mechanism of barite nucleation and growth without and with the influence of Sr^{2+}.

the absorption of Ba^{2+} to a surface (Figure 8).[31] This is justified because our experiments were conducted in pH = 7 aqueous solutions, where the silica surfaces are likely to be negatively charged[49,50] and will disfavor adsorption of negatively charged SO_4^{2-} ions. In the case where we have Ba^{2+}:Sr^{2+} = 1:1, the total number of divalent metal cations in the system is doubled, and if we assume that Ba^{2+} and Sr^{2+} have similar affinities for the silica surface, the total number of adsorbed cations on the silica surface likely increases. Previous studies have shown that the hydration behavior of cations is the key mechanism governing their absorption on silicates and subsequent crystal growth in general.[31,51,52] Both Sr^{2+} and Ba^{2+} are divalent ions, and both Sr^{2+} and Ba^{2+} have similar large, exothermic hydration energies ($\Delta H_{hydration}$ = −1443 kJ/mol for Sr^{2+} and −1305 kJ/mol for Ba^{2+}).[53] Thus, it is likely that Sr^{2+} and Ba^{2+} have similar affinities for silica surfaces, and indeed previous crystal truncation rod measurements indicate Sr^{2+} can form both inner- and outer-sphere complexes upon absorption to silicate minerals,[54] and the adsorption Sr^{2+} and Ba^{2+} on silica have been found to have similar equilibrium constants for adsorption.[55] These relatively stable adsorbed Sr^{2+} species could provide structurally similar precursor sites for the subsequent absorption of SO_4^{2-} and/or other reactions leading to the formation of stable barite nuclei. A supporting argument for this interpretation is that the pre-factor, rather than the interfacial energy, was found to be responsible for the increased nucleation rates in the Sr-rich solutions relative to the Sr-free case. If the strontium were affecting the nucleation mechanism directly, one would expect a larger change in the interfacial free energy in the Sr-rich solution. What we envision is a multistep nucleation process, whereby ions such as Sr^{2+} and Ba^{2+} first adsorb to silica surfaces, and subsequent nucleation processes occur using these as precursor sites. Thus, the concentration of adsorbed species should be a variable in the CNT expression (eq 1), reflected here as changes in the pre-factor.

As a further note, previous work by Dai et al.[31] could not detect Ba^{2+} adsorption onto bare glass using a quartz crystal microbalance at SI = 1.1, while significant adsorption was found on organic-modified surfaces coated with self-assembled monolayers (SAM). However, an increase in the amount of absorbed Ba^{2+} (or Sr^{2+})[31] on a SAM did not necessarily correlate with a higher nucleation rate of barite, indicating there are interfacial processes other than an initial cation adsorption that limit the rate of nucleation. Since, in our case, a higher nucleation rate is observed, it is likely that the effects of strontium on barite nucleation are also substrate-dependent (silica vs SAM). We speculate that the origin of this substrate dependence is because the mechanisms of nucleation change on a SAM relative to a silica surface. This is justified because nucleation rates on SAMs have been observed to be orders of magnitude higher than that on the silica used here. This argument is supported by observations that the orientations of nuclei can change depending on the terminal moiety of a SAM.[56,57]

Strontium-Inhibited Barite Growth. Impurities are known to change growth rates and growth habits of crystals.[58−61] Impurity ions adsorbed at surface sites, including monomolecular steps or kink sites on those steps result in retardation of the step advance. Both adsorption of Sr^{2+} on barite and precipitation of a $(Ba,Sr)SO_4$ solid solution

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Cryst. Growth Des. 2021, 21, 5828−5839

could decrease the growth rate. Weber et al.[11] showed that the barite growth rate decreased at the 1:1 aqueous Ba^{2+}:Sr^{2+} ratios used here. High-resolution chemical imaging showed uniform incorporation of Sr on barite surface steps. X-ray reflectivity found that Sr^{2+} can simultaneously incorporate and adsorb on the barite (001) surface.[62] Formation of $(Ba,Sr)SO_4$ solid solutions of any composition could lower the SI and, therefore, the growth rate (Figure 7a). Even if Sr^{2+} did not incorporate into barite, inhibition of barite growth could be achieved by non-incorporating impurity effects through influencing compositions of growing surfaces.[63] Our substrate surface area-based growth rates are similar in both Sr-free and Sr-bearing solutions (Table 3), reflecting the same chemical potential that provides the free energy gain for crystal growth. Addition of Sr^{2+} inhibited barite growing into large particles, yielding more nuclei and higher crystal surface area, resulting in a reduction of crystal surface area-based growth rate (Figure 8).

CONCLUSIONS

We investigated the role of Sr^{2+} in the heterogeneous nucleation of barite on silica surfaces and compared classical optical microscopy with advanced XnT. Optical microscopy failed to resolve small nuclei in the early stage, resulting in a lower nucleation rate than the results obtained by XnT. Addition of Sr^{2+} was found to promote the nucleation rate by primarily increasing the kinetic pre-factor in CNT. To explain this, we suggest that the adsorbed strontium species serve as precursor sites for nucleation in a similar capacity to the adsorbed barium ions but without changing the mechanism of nucleation. Crystals with different orientations exhibited different incubation times in Sr-free solutions. Crystals with their (001) surface attached appeared later on the substrate than crystals attached by a high-index surface. The comparison of both imaging methods further improved our understanding of the nucleation and growth mechanism of barite and reflected a counter effect of impurity metals on the crystal nucleation and growth kinetics.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.cgd.1c00715.

> Solution chemistry; liquid XnT flow cell and error analysis; additional optical and nanotomography images; calculated parameters for solid solutions in CNT; growth rates obtained from optical images; growth rates calculated using surface mesh areas; changes in pre-factor and interfacial energy to the calculated nucleation rates (PDF)

AUTHOR INFORMATION

Corresponding Authors

Ke Yuan − *Chemical Sciences Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States;* orcid.org/0000-0003-0565-0929; Email: yuank@ornl.gov

Andrew G. Stack − *Chemical Sciences Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States;* orcid.org/0000-0003-4355-3679; Email: stackag@ornl.gov

Authors

Vitalii Starchenko − *Chemical Sciences Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States;* orcid.org/0000-0003-2313-7379

Nikhil Rampal − *Chemical Sciences Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; Department of Chemical Engineering, Columbia University, New York, New York 10027, United States*

Fengchang Yang − *Chemical Sciences Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States*

Xiaogang Yang − *Center for X-ray and Nano Science, Deutsches Elektronen-Synchrotron DESY, 22607 Hamburg, Germany*

Xianghui Xiao − *National Synchrotron Light Source-II, Brookhaven National Laboratory, Upton, New York 11973, United States*

Wah-Keat Lee − *National Synchrotron Light Source-II, Brookhaven National Laboratory, Upton, New York 11973, United States*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.cgd.1c00715

Author Contributions

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences, Chemical Sciences, Geosciences, and Biosciences Division. This research used resources at the FXI beamline (18-ID) of the National Synchrotron Light Source II, a U.S. Department of Energy (DOE) Office of Science User Facility operated for the DOE Office of Science by Brookhaven National Laboratory under Contract No. DE-SC0012704. This research was conducted, in part, using instrumentation within ORNL's Materials Characterization Core provided by UT-Battelle, LLC under Contract No. DE-AC05-00OR22725 with the U.S. Department of Energy. This research used resources of the Compute and Data Environment for Science (CADES) at the Oak Ridge National Laboratory, which is supported by the Office of Science of the U.S. Department of Energy under Contract No. DE-AC05-00OR22725.

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https://doi.org/10.1021/acs.cgd.1c00715
Cryst. Growth Des. 2021, 21, 5828−5839